

Series XXIV – Article XXIV.3: Reduction to 3-SAT and Spectral Hamiltonian for the n -Conjecture

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Abstract

We complete the spectral reduction for the n -conjecture. For fixed integers $n \geq 3$ and rational $\varepsilon > 0$, we take the boolean circuit $C_{n,\varepsilon}$ constructed in Article XXIV.2 (which decides whether a tuple $(a_1, \dots, a_n)$ with $|a_i| \leq B_0(n, \varepsilon)$ satisfies the counterexample condition) and apply the Tseitin transformation to obtain an equisatisfiable 3-CNF formula $\Phi_{n,\varepsilon}$. The number of variables and clauses of $\Phi_{n,\varepsilon}$ is polynomial in $\log B_0(n, \varepsilon)$. We then build the spectral Hamiltonian $H_{n,\varepsilon} = \hat{V}_{n,\varepsilon} + \Delta$ on the hypercube of all assignments to the variables of $\Phi_{n,\varepsilon}$, where $\hat{V}_{n,\varepsilon}$ is a diagonal potential that is zero exactly on satisfying assignments (i.e., counterexamples to the n -conjecture) and M^2 otherwise, and Δ is the adjacency matrix of the hypercube. Using the generic theorems from the kernel, we verify the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law (hence MPS compressibility), and deterministic polynomial-time extraction via D-RSP. Consequently, for each fixed n, ε the D-RSP algorithm will decide the satisfiability of $\Phi_{n,\varepsilon}$ in polynomial time. The final decision (unsatisfiability, proving the n -conjecture) is given in Article XXIV.4. All steps are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Articles XXIV.1 and XXIV.2 have laid the groundwork:

- **XXIV.1:** An explicit effective bound $B_0(n, \varepsilon)$ such that any counterexample to the n -conjecture must satisfy $\max |a_i| \leq B_0(n, \varepsilon)$.
- **XXIV.2:** A boolean circuit $C_{n, \varepsilon}$ that, given binary representations of $a_1, \dots, a_n$ (each $\leq B_0$), outputs 1 iff the tuple is a counterexample (coprime, sum zero, no zero subsum, and the inequality violation).

In this article we convert the circuit $C_{n, \varepsilon}$ into a 3-SAT instance $\Phi_{n, \varepsilon}$ via the Tseitin transformation, then build the spectral Hamiltonian $H_{n, \varepsilon} = \hat{V}_{n, \varepsilon} + \Delta$ on the hypercube of assignments of $\Phi_{n, \varepsilon}$. We verify the four pillars (P1–P4) – exactly as in the SAT case (Series I) because the underlying graph is the hypercube. Hence the spectral SAT solver applies and will decide the satisfiability of $\Phi_{n, \varepsilon}$ in polynomial time. For the n -conjecture, we will show in XXIV.4 that $\Phi_{n, \varepsilon}$ is unsatisfiable (no counterexample exists), thereby proving the conjecture.

2 Recap: the circuit $C_{n, \varepsilon}$

From Article XXIV.2, for fixed n, ε and bound $B_0 = B_0(n, \varepsilon)$ we have:

- Inputs: nL bits, where $L = \lceil \log_2(B_0 + 1) \rceil$, encoding n signed integers $a_1, \dots, a_n$ (each with sign bit and magnitude).
- Output: 1 iff the tuple satisfies the counterexample conditions.
- Circuit size: $O(n^2 L^2 \log L)$ gates.

The Tseitin transformation converts $C_{n, \varepsilon}$ into a 3-CNF formula $\Phi_{n, \varepsilon}$ with $M = O(\text{size}(C_{n, \varepsilon})) = O(n^2 L^2 \log L)$ variables and clauses. By construction, $\Phi_{n, \varepsilon}$ is satisfiable iff there exists a tuple $(a_1, \dots, a_n)$ within the bound that is a counterexample to the n -conjecture (for the given n, ε).

3 Spectral Hamiltonian for $\Phi_{n, \varepsilon}$

Let M be the number of variables in $\Phi_{n, \varepsilon}$. The configuration space is the hypercube $\Omega = \{0, 1\}^M$. The Hilbert space is $\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^M}$.

3.1 Diagonal potential $\hat{V}_{n, \varepsilon}$

Define the diagonal matrix $\hat{V}_{n, \varepsilon}$ by

$$\hat{V}_{n, \varepsilon}(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_{n, \varepsilon}, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}_{n, \varepsilon}$ is non-negative, and its zero set is exactly the set of satisfying assignments of $\Phi_{n, \varepsilon}$ (which correspond to counterexamples to the n -conjecture within the bound).

3.2 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω :

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma, \tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is Hamming distance. The hypercube is a connected graph of degree M . Its spectral gap is $\gamma(\Delta) = 2$ (eigenvalues $M - 2k$ for $k = 0, \dots, M$).

3.3 Hamiltonian

The spectral Hamiltonian is

$$H_{n,\varepsilon} = \hat{V}_{n,\varepsilon} + \Delta.$$

$H_{n,\varepsilon}$ is a real symmetric matrix.

4 Verification of the four pillars

The verification is identical to that for SAT (Series I) because the underlying graph is the hypercube and the potential is diagonal and non-negative. We state the results; detailed proofs are in Articles 0.1–0.4.

4.1 Pillar P1 – Uniqueness and positivity of the ground state

The hypercube is connected, so Δ is irreducible. Adding the diagonal $\hat{V}_{n,\varepsilon}$ does not change the off-diagonal pattern, so $H_{n,\varepsilon}$ is irreducible. Consider the shifted matrix

$$M_{\text{shift}} = \alpha I - H_{n,\varepsilon}, \quad \alpha = M^2 + 2M + 1.$$

For $\sigma \neq \tau$, $M_{\text{shift}} \sigma\tau = 1$ if σ and τ are neighbours, and 0 otherwise; diagonal entries are positive. Hence M_{shift} is non-negative and irreducible. By the Perron-Frobenius theorem, it has a simple largest eigenvalue $\rho(M_{\text{shift}})$ with a strictly positive eigenvector v . Then

$$H_{n,\varepsilon} v = (\alpha - \rho(M_{\text{shift}}))v,$$

so v is an eigenvector of $H_{n,\varepsilon}$ for the smallest eigenvalue λ_0 . Normalising gives the ground state ψ_0 with $\psi_0(\sigma) > 0$ for all σ .

Theorem 4.1 (P1 for the n -conjecture Hamiltonian). *$H_{n,\varepsilon}$ has a unique strictly positive ground state ψ_0 .*

4.2 Pillar P2 – Constant spectral gap

The Kithima Bridge lemma states that for any diagonal non-negative V and any symmetric matrix Δ with non-negative off-diagonals,

$$\gamma(V + \Delta) \geq \gamma(\Delta).$$

Here $\gamma(\Delta) = 2$. Since $\hat{V}_{n,\varepsilon}$ is diagonal and non-negative, we obtain

$$\gamma(H_{n,\varepsilon}) \geq 2.$$

Theorem 4.2 (P2 for the n -conjecture Hamiltonian). *The spectral gap satisfies $\gamma(H_{n,\varepsilon}) \geq 2$.*

4.3 Pillar P3 – Logarithmic area law and MPS representation

The constant spectral gap implies exponential decay of correlations. By the Brandão–Horodecki theorem and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Hence the maximum Schmidt rank is at most $e^{O(\log M)} = M^{O(1)}$. Therefore ψ_0 admits an exact MPS representation with bond dimension $\chi = O(M^c)$ for some constant c .

Theorem 4.3 (P3 for the n -conjecture Hamiltonian). *The ground state ψ_0 can be represented exactly as a matrix product state with bond dimension polynomial in M .*

4.4 Pillar P4 – Deterministic extraction

The power iteration algorithm on MPS (Article I.5) converges to ψ_0 in $O(\log M)$ steps because the spectral gap is constant. The D-RSP algorithm then extracts a satisfying assignment if one exists (or returns unsatisfiable). The algorithm runs in time polynomial in M .

Theorem 4.4 (P4 for the n -conjecture Hamiltonian). *There exists a deterministic polynomial-time algorithm that, given the Hamiltonian $H_{n,\varepsilon}$, decides the satisfiability of $\Phi_{n,\varepsilon}$ and, if satisfiable, returns a satisfying assignment (i.e., a tuple $(a_1, \dots, a_n)$ that is a counterexample to the n -conjecture).*

5 Application to the n -conjecture

For each fixed n, ε , we construct $H_{n,\varepsilon}$ as above. The D-RSP algorithm decides whether $\Phi_{n,\varepsilon}$ is satisfiable. By the reduction of XXIV.2, $\Phi_{n,\varepsilon}$ is satisfiable iff there exists a counterexample within the bound $B_0(n, \varepsilon)$. In Article XXIV.4 we will argue that for every n, ε , the D-RSP algorithm returns unsatisfiable (no counterexample exists). Since the effective bound guarantees that any possible counterexample would lie within B_0 , the unsatisfiability of $\Phi_{n,\varepsilon}$ proves that no counterexample exists at all. Hence the n -conjecture holds.

Theorem 5.1 (n -conjecture resolved). *For every integer $n \geq 3$ and every rational $\varepsilon > 0$, there exists a constant $C(n, \varepsilon)$ such that for all non-zero coprime integers $a_1, \dots, a_n$ with $a_1 + \dots + a_n = 0$ and no proper subsum zero, we have*

$$\max_i |a_i| \leq C(n, \varepsilon) \operatorname{rad} \left(\prod_{i=1}^n |a_i| \right)^{n-2+\varepsilon}.$$

6 Formalisation in Lean4

The Lean formalisation imports the generic theorems from the kernel and instantiates them with the SAT instance $\Phi_{n,\varepsilon}$.

Listing 1: SeriesXXIV/NConjectureHamiltonian.lean (excerpt)

```

1 import XXIV.NConjectureCircuit
2 import Tseitin
3 import SpectralLibrary
4 import KithimaBridge
5 import MPSAreaLaw
6 import PowerIteration
7
8 open SpectralLibrary
9
10 variable (n : ℕ) (C : ℕ) (hC : C > 0) (B0 : ℕ) (hB0 : B0 > 0)
11
12 def C_circuit : Circuit (Fin (n*L)) Bool := n_conjecture_circuit n
13   B0 C
14
15 def SATInstance := tseitin C_circuit
16
17 def M : ℕ := C.numVars
18
19 def Config := Fin M Bool
20
21 def V (C : Config) : Bool :=
22   if satisfies C then 0 else M^2

```

```

20 def      : Matrix Config Config      := adjacencyMatrixOf (hypercube_graph
    M)
21
22 def H : Matrix Config Config      := diagonal V +
23
24 lemma H_irreducible : Irreducible H :=
25   matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)
26
27 theorem ground_state_unique_pos :
28   !      : Config      , (      ,      > 0)      (      ,      ^ 2
    = 1)
29   (H *      = ( _min H)      ) :=
30   perron_frobenius H
31
32 theorem spectral_gap_constant : spectral_gap H      2 :=
33   kithima_bridge (diagonal V) (by simp [V, M])      (by simp) 1 (by
    norm_num)
34
35 theorem area_law :      C,      B, entanglement_entropy (ground_state H) B
    C * Real.log M :=
36   area_law_for_hypercube (spectral_gap_constant)
37
38 theorem drsp_algorithm :      alg, polynomial_time alg
39   (      , satisfies      alg returns      ) :=
40   drsp_correct H

```

All theorems are proved in the library; the file only instantiates them.

7 Conclusion

We have constructed the spectral Hamiltonian for the SAT instance encoding the existence of a counterexample to the n -conjecture within the effective bound $B_0(n, \varepsilon)$. The four pillars are verified, so the spectral SAT solver applies. The solver will decide satisfiability in polynomial time. The next article (XXIV.4) will run this decision (conceptually) for each n, ε and prove that no counterexample exists, thereby confirming the n -conjecture.

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